

Web Platform for Long-Range Weather Data Analysis

Supervisor: **Dr. Zakharia Frenkel**

Presenting: **Idan Aizenberg | Aviv Aytek**

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Background and Problem

Long-range weather data is critical for decision-making across multiple industries.

However, existing solutions struggle with efficient data access, flexible querying, and clear visualization of large spatio-temporal datasets.

WeatherSight provides a unified, scalable platform for searching, analyzing, and visualizing long-range weather data, while enabling weather forecasting companies to integrate and present their own datasets through the platform.

Requirements and Goals

- Support location-based weather search using geographic coordinates
- Enable parameter-based search across multiple weather variables
- Allow advanced filtering by date ranges and thresholds
- Visualize results using charts, maps, and comparative views

Challenges and Solutions

- Slow geographic proximity searches → KD-tree-based spatial indexing
- Complex multi-parameter queries → Dedicated API routes and query optimization
- High query load and repeated requests → Caching and usage tracking
- Balancing flexibility and usability → Structured, intuitive search interface

System Overview

User Interaction (Frontend)

- Users perform location-based and parameter-based searches through a web interface.

Geocoding & Spatial Search

- Location names are converted to coordinates, and a KD-tree is used to find the nearest weather grid point.

API Layer

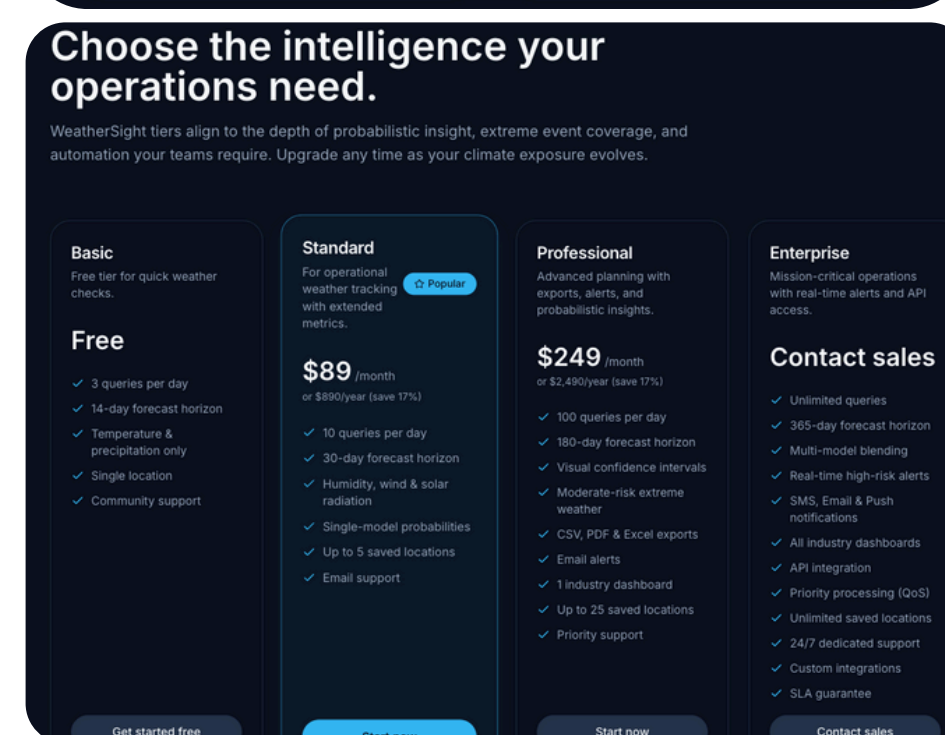
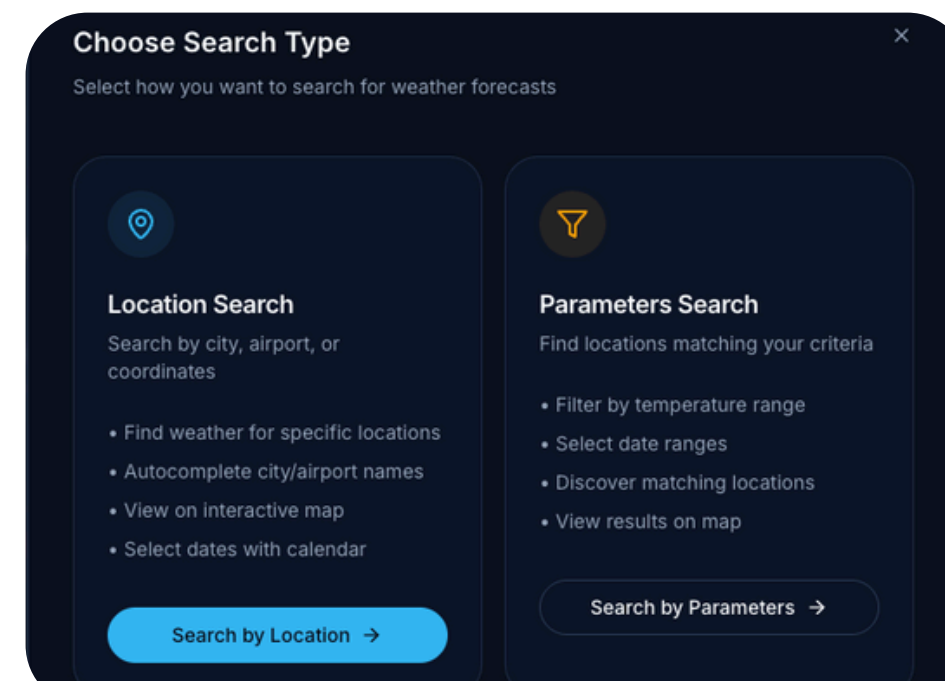
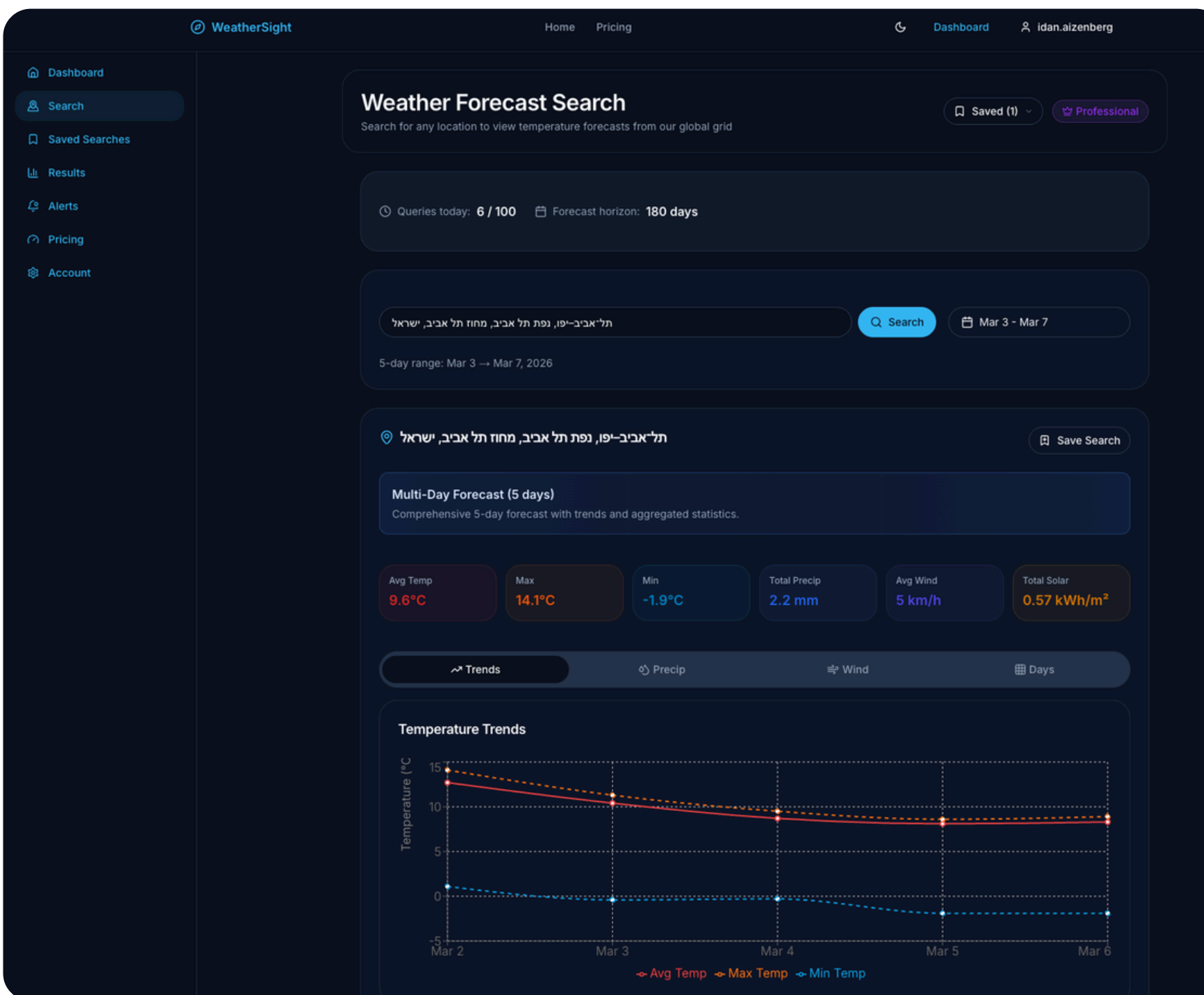
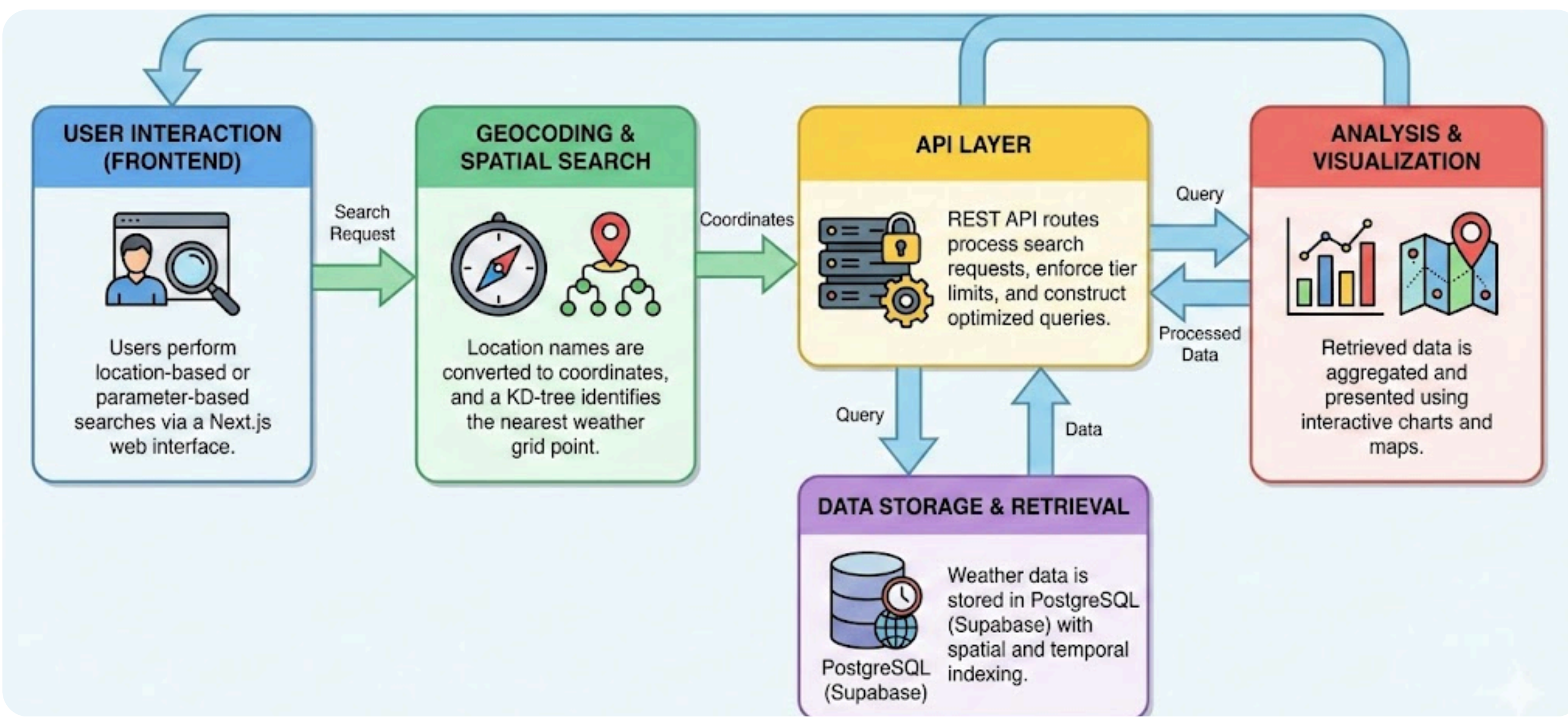
- REST API routes process requests, enforce subscription tiers, and optimize queries.

Data Storage & Retrieval

- Weather data is stored in PostgreSQL (Supabase) with spatial and temporal indexing.

Analysis & Visualization

- Retrieved data is aggregated and presented using interactive charts and maps.



Technologies Used

Frontend

- Next.js ,TypeScript, Tailwind CSS
- Recharts, Leaflet

Backend & Database

- Supabase (PostgreSQL), RESTful API

Data Processing

- KD-tree spatial indexing, Geocoding services
- Time-series and geospatial queries

Dev & Infrastructure

- Git & GitHub



Summary and Future Work

WeatherSight demonstrates how advanced data structures, optimized database design, and modern web technologies can be combined to create an efficient platform for long-range weather data analysis and visualization.

Future work will include:

- Machine learning-based weather pattern analysis
- Real-time alerts and notifications
- Advanced caching and performance optimization
- Industry-specific analytical dashboards